

Cloud to KVM Repatriation

AWS, Azure & GCP Exit Deep-Dive

Cloud costs grow 15-25% annually while on-premises hardware costs decrease. Organizations are repatriating workloads from AWS, Azure, and GCP to KVM infrastructure. This guide covers VM export from every major cloud provider, image conversion, cost modeling, and hybrid landing zone architecture.

AWS AMI Export | Azure VHD | GCP Image | Cost Modeling | Hybrid | 10 Cloud Sources | \$1.8M Savings

Cloud Cost Reality Check

Why organizations are bringing workloads back on-premises.

Cloud vs On-Prem 3-Year TCO (100 VMs)

Cost Component	AWS	Azure	GCP	On-Prem KVM	Notes
Compute (3yr)	\$1,080,000	\$1,020,000	\$960,000	\$180,000	On-prem: server hardware amortized
Storage (50TB, 3yr)	\$138,000	\$132,000	\$120,000	\$25,000	On-prem: NVMe SSDs, one-time purchase
Network egress (3yr)	\$216,000	\$180,000	\$144,000	\$0	Cloud egress: \$0.08-\$0.12/GB
Software licenses	\$0	\$0	\$0	\$0	KVM: free, Apache-2.0
Operations (staff, 3yr)	\$360,000	\$360,000	\$360,000	\$360,000	Same staffing either way
Data center (colo, 3yr)	\$0	\$0	\$0	\$108,000	4 racks, power, cooling
Migration (one-time)	—	—	—	\$45,000	hyper2kvm + labor
3-Year Total	\$1,794,000	\$1,692,000	\$1,584,000	\$718,000	
Savings vs Cloud				\$866K-\$1.08M	55-60% reduction

When to Repatriate

- Steady-state workloads (predictable, not bursty)
- Data-heavy applications (storage + egress costs dominate)
- Compliance requirements (data residency, air-gap)
- Performance-sensitive (latency, GPU, bare-metal needs)
- Cloud bill growing > 20% annually without new workloads
- Multi-year commitments expiring (RI/savings plans ending)

When to Stay in Cloud

- Highly variable workloads (autoscaling critical)
- Global presence needed (30+ regions)
- Managed services dependency (RDS, Lambda, etc.)
- Small footprint (< 20 VMs — cloud is simpler)
- No DC operations expertise on staff
- Short-lived workloads (< 12 months)

AWS VM Export

Exporting EC2 instances and AMIs for conversion to KVM.

AWS Export Methods

Method 1: VM Export (create-instance-export-task)

```
# Export running EC2 instance to S3
aws ec2 create-instance-export-task \
  --instance-id i-0123456789abcdef0 \
  --target-environment vmware \
  --export-to-s3-task \
    DiskImageFormat=vmdk,\
    ContainerFormat=ova,\
    S3Bucket=my-export-bucket,\
    S3Prefix=exports/

# Monitor export progress
aws ec2 describe-export-tasks \
  --export-task-ids export-i-0123456789

# Download from S3
aws s3 cp s3://my-export-bucket/exports/export-i-012.ova ./

# Convert with h2kvmctl
h2kvmctl convert --source export.ova \
  --format qcow2 --fix-all --deploy-libvirt
```

Method 2: EBS Snapshot → Raw Image

```
# Create snapshot of EBS volume
aws ec2 create-snapshot \
  --volume-id vol-0123456789abcdef0 \
  --description "Repatriation export"

# Share snapshot to S3 (via EBS direct API)
aws ebs get-snapshot-block \
  --snapshot-id snap-0123456789 \
  --block-index 0 --block-token xxx

# Or: attach snapshot to helper EC2, dd to S3
# Faster for large disks
aws ec2 create-volume --snapshot-id snap-012 \
  --availability-zone us-east-1a
# Attach to helper → dd if=/dev/xvdf | gzip | \
#   aws s3 cp - s3://bucket/disk.raw.gz

# Convert raw to QCOW2
h2kvmctl convert --source disk.raw \
  --format qcow2 --fix-all
```

AWS-Specific Fixes

Issue	Cause	Fix
NVMe device names (/dev/nvme0n1)	AWS Nitro instances use NVMe	--fix-fstab converts to UUID (works for both)
cloud-init conflicts	cloud-init resets network on boot	Disable: touch /etc/cloud/cloud-init.disabled
AWS SSM agent running	SSM agent can't reach AWS endpoints	Uninstall: dnf remove amazon-ssm-agent
Elastic Network Adapter (ENA) driver	ENA driver not needed on KVM	Removed automatically; virtio-net used instead

IMDSv2 dependency

Apps querying 169.254.169.254

Update apps to use local config instead of IMDS

Azure VM Export

Exporting Azure VMs as VHD/VHDX for conversion to KVM.

Azure Export Workflow

```
# Step 1: Deallocate VM (required for export)
az vm deallocate --resource-group myRG --name my-vm

# Step 2: Generalize (optional – skip if preserving identity)
# az vm generalize --resource-group myRG --name my-vm

# Step 3: Get SAS URL for managed disk download
DISK_ID=$(az vm show -g myRG -n my-vm --query "storageProfile.osDisk.managedDisk.id" -o tsv)
az disk grant-access --duration-in-seconds 86400 --ids $DISK_ID --query accessSas -o tsv
# Returns: https://md-xxx.blob.core.windows.net/xxx?sv=...&sig=...

# Step 4: Download VHD (use azcopy for speed)
azcopy copy "https://md-xxx.blob.core.windows.net/xxx?sv=...&sig=..." ./my-vm.vhd

# Step 5: Convert VHD to QCOW2 with h2kvmctl
h2kvmctl convert --source my-vm.vhd --format qcow2 \
  --fix-fstab --fix-initramfs --fix-network \
  --emit-domain-xml --deploy-libvirt

# Step 6: Revoke SAS access
az disk revoke-access --ids $DISK_ID
```

Azure-Specific Fixes

Linux VMs from Azure

- **waagent:** Azure Linux Agent — disable/remove
- `systemctl disable walinuxagent && dnf remove WALinuxAgent`
- **cloud-init Azure datasource:** disable
- `rm /etc/cloud/cloud.cfg.d/10_azure.cfg`
- **Hyper-V drivers:** `hv_vmbus`, `hv_storvsc`, `hv_netvsc`
- `h2kvmctl` replaces with virtio drivers in `initramfs`
- **Swap:** Azure creates swap on resource disk — configure locally
- **NTP:** Remove Azure-specific chrony config

Windows VMs from Azure

- **Azure Guest Agent:** uninstall from Programs
- **Hyper-V IC:** Integration Components stay (harmless)
- **VirtIO drivers:** inject with `h2kvmctl --virtio-drivers`
- **Activation:** Azure KMS won't work on-prem
- Re-activate with on-prem KMS or MAK key
- **RDP:** works immediately (same settings preserved)
- **Azure AD:** if joined, may need to rejoin on-prem AD
- **Disk format:** Azure exports as fixed VHD (fast convert)

GCP VM Export

Exporting Google Compute Engine instances for KVM conversion.

GCP Export Methods

Method 1: gcloud export (Recommended)

```
# Export VM disk to GCS as VMDK
gcloud compute images export \
  --destination-uri gs://my-bucket/exports/vm-disk.vmdk \
  --image my-vm-image \
  --export-format vmdk

# Or export as raw (faster, larger)
gcloud compute images export \
  --destination-uri gs://my-bucket/exports/vm-disk.raw.tar.gz \
  --image my-vm-image \
  --export-format raw

# Download from GCS
gsutil cp gs://my-bucket/exports/vm-disk.vmdk ./

# Convert
h2kvmctl convert --source vm-disk.vmdk \
  --format qcow2 --fix-all --deploy-libvirt
```

Method 2: Disk Snapshot → Image → Export

```
# Create snapshot of running VM's disk
gcloud compute disks snapshot my-vm-disk \
  --snapshot-names my-vm-snap \
  --zone us-central1-a

# Create image from snapshot
gcloud compute images create my-vm-image \
  --source-snapshot my-vm-snap

# Export image (as above)
gcloud compute images export \
  --destination-uri gs://bucket/vm.vmdk \
  --image my-vm-image \
  --export-format vmdk

# Clean up
gcloud compute images delete my-vm-image
gcloud compute snapshots delete my-vm-snap
```

All Cloud Providers: Export Comparison

Cloud	Export Format	Export Method	Speed	Egress Cost	Notes
AWS	VMDK, VHD, RAW	export-task → S3	Medium	\$0.09/GB	Requires vmimport role; limited to EBS-backed
Azure	VHD (fixed)	SAS URL → download	Fast	\$0.087/GB	Must deallocate VM first; azcopy recommended
GCP	VMDK, RAW	image export → GCS	Fast	\$0.12/GB	Uses Daisy workflow; can export running disk snapshot
Oracle Cloud	QCOW2	object storage export	Medium	\$0.0085/GB	Cheapest egress; native QCOW2 export
DigitalOcean	VMDK, QCOW2, RAW	snapshot → spaces	Fast	\$0.01/GB	Custom image export via API

Linode/Akamai	RAW (dd)	Rescue mode → dd	Slow	Free (1TB/mo)	Manual process, requires rescue boot
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Cloud-Specific Driver Cleanup

Removing cloud agents and replacing cloud drivers with KVM VirtIO.

Agent & Driver Removal Checklist

Cloud	Agent/Driver	Remove Command	Safe to Remove?	Notes
AWS	amazon-ssm-agent	<code>dnf remove amazon-ssm-agent</code>	Yes	Systems Manager agent, no function outside AWS
AWS	aws-cli (if installed)	Keep if used locally	Optional	Can still use AWS CLI for S3/API access from on-prem
AWS	ENA driver (ena.ko)	Removed with <code>initramfs rebuild</code>	Yes	Replaced by virtio-net
Azure	WALinuxAgent	<code>dnf remove WALinuxAgent</code>	Yes	Azure provisioning agent, remove completely
Azure	Hyper-V IC (hv_*)	Keep (harmless)	Optional	Kernel modules, won't load without Hyper-V bus
Azure	Azure Guest Agent (Win)	Uninstall from Programs	Yes	Windows agent, remove via Control Panel
GCP	google-guest-agent	<code>dnf remove google-guest-agent</code>	Yes	GCP metadata agent, safe to remove
GCP	google-osconfig-agent	<code>dnf remove google-osconfig-agent</code>	Yes	OS Config management, not needed on-prem
All	cloud-init	<code>touch /etc/cloud/cloud-init.disabled</code>	Yes	Disable (don't remove) to prevent network reset on boot

h2kvmctl Cloud Cleanup

```
# h2kvmctl auto-detects cloud source and cleans up
h2kvmctl convert --source aws-vm.vmdk --format qcow2 --fix-all

# Automatic cleanup includes:
# 1. Detect cloud provider from installed agents/drivers
# 2. Disable cloud-init (preserve config for reference)
# 3. Remove cloud-specific network rules (169.254.x.x routes)
```

```
# 4. Rebuild initramfs with virtio drivers (removes cloud drivers)
# 5. Fix fstab (NVMe → virtio device names via UUID)
# 6. Update GRUB (remove cloud-specific console parameters)
# 7. Configure serial console for KVM (ttyS0)
# 8. Set timezone to local (was UTC for cloud billing)
```

Hybrid Landing Zone Architecture

Not everything needs to come back — design a hybrid architecture that optimizes cost and capability.

Workload Placement Decision Matrix

Workload Type	Keep in Cloud	Repatriate	Rationale	Example
Steady-state databases		Repatriate	Predictable, storage-heavy, high egress	PostgreSQL, Oracle, SQL Server
Burstable web apps	Cloud		Autoscaling critical, global CDN needed	Marketing site, SaaS frontend
AI/ML training		Repatriate	GPU costs 3x lower on-prem	LLM training, model fine-tuning
ML inference (global)	Cloud		Low latency to users worldwide	Real-time prediction API
CI/CD pipelines		Repatriate	Predictable compute, data locality	Jenkins, GitLab CI runners
Dev/test environments	Cloud		Ephemeral, on-demand, cost-effective spot	Feature branch environments
Backup / DR	Cloud (S3)		Cheap storage, global availability	Restic to S3, Velero backup
ERP / LOB apps		Repatriate	Steady state, data-sensitive, compliance	SAP, healthcare EHR

Hybrid Architecture

On-Premises (KVM)

- Databases: PostgreSQL, MySQL, Oracle, SQL Server
- Application servers: Java, .NET, Python backends
- AI/ML training: GPU clusters with passthrough
- CI/CD: Jenkins/GitLab runners, artifact storage
- Internal services: AD, DNS, DHCP, monitoring
- Compliance workloads: HIPAA, PCI, SOX

Cloud (AWS/Azure/GCP)

- Web frontends: CDN, edge, autoscaling
- Managed services: SQS, SNS, Pub/Sub, Lambda
- Backup: S3/GCS object storage (\$0.023/GB/mo)
- DR: standby VMs for site failover
- Dev/test: ephemeral, spot instances
- SaaS connectors: API gateways, webhooks

Egress Cost Optimization

Minimizing the one-time cost of downloading VMs from cloud providers.

Egress Cost for 100 VMs (10 TB total)

Provider	Rate	10 TB Cost	Free Tier	Optimization Strategy
AWS	\$0.09/GB	\$900	100 GB/mo	Use AWS Snowball (\$300 flat) for > 10 TB; S3 Transfer Acceleration
Azure	\$0.087/GB	\$870	100 GB/mo	Azure Data Box (\$500 flat) for > 10 TB; azcopy with compression
GCP	\$0.12/GB	\$1,200	1 GB/mo	Google Transfer Appliance for > 20 TB; compress before transfer
Oracle Cloud	\$0.0085/GB	\$85	10 TB/mo	10 TB free egress/month — export over 1 month for free
DigitalOcean	\$0.01/GB	\$100	1 TB/mo	Very cheap; use bandwidth pool across droplets

Transfer Optimization

Compress Before Download

```
# Compress VM image in cloud before egress
# Saves 40-60% on egress costs

# AWS: compress in EC2 before S3
ssh ec2-helper "qemu-img convert -c \
  -f raw -O qcow2 /dev/xvdf disk.qcow2"
# Upload compressed to S3, download compressed

# Typical compression ratios:
# Linux VM: 60-70% reduction
# Windows VM: 40-50% reduction
# Database VM: 50-60% reduction
# Empty space: 95%+ reduction

# 10 TB raw → 4 TB compressed → $360 egress (vs $900)
```

Physical Transfer Devices

- **AWS Snowball Edge:** 80 TB, \$300 per job
- **Azure Data Box:** 100 TB, \$500 per device
- **Google Transfer Appliance:** 300 TB, custom pricing
- **Break-even:** physical device cheaper at ~5 TB+
- Ship time: 5-10 business days typical
- Encrypted in transit: AES-256 hardware encryption
- Best for: large migrations (50+ TB), slow internet

Implementation & ROI Summary

Phase-by-phase repatriation plan and financial impact.

Repatriation Timeline (100 VMs from AWS)

Phase	Duration	Activity	Deliverable
1. Assessment	2 weeks	Cloud cost analysis, workload classification, hybrid architecture design	Repatriation plan, workload placement matrix, cost model
2. Infrastructure	3 weeks	Procure/deploy on-prem servers, install KVM, configure networking	Production-ready KVM cluster (storage, network, HA)
3. Export	2 weeks	Export VMs from AWS (compress, download, verify checksums)	100 VM images on local storage, verified integrity
4. Convert	1 week	h2kvmctl batch conversion, fix drivers/fstab/initramfs, cloud cleanup	100 QCOW2 images + libvirt domain XMLs
5. Test	2 weeks	Boot test, app validation, performance baseline, DR setup	Test results, performance comparison, DR verified
6. Cutover	2 weekends	DNS cutover, go-live per wave, monitor, validate	All VMs running on-prem, cloud instances stopped
7. Decommission	1 month	Monitor for 30 days, terminate cloud instances, close accounts	Cloud spend reduced to \$0 (or hybrid target)

Financial Summary

One-Time Migration Costs

- On-prem hardware (8 servers): \$240,000
- Network equipment (switches, firewall): \$30,000
- Colocation setup (4 racks): \$15,000
- Migration labor (12 weeks): \$60,000
- Cloud egress (10 TB compressed): \$400
- hyper2kvm software: \$0 (Apache-2.0)
- Total one-time: \$345,400**

Annual Ongoing Savings

- Cloud compute eliminated: \$360,000/yr
- Cloud storage eliminated: \$46,000/yr
- Cloud egress eliminated: \$72,000/yr
- New on-prem colo cost: -\$36,000/yr
- Power/cooling: -\$18,000/yr
- Net annual savings: \$424,000/yr**
- Payback period: 9.8 months**

Cloud Repatriation Pays for Itself in Under 10 Months

100 VMs repatriated from AWS to KVM saves \$424K/yr with a \$345K one-time investment — breakeven in 9.8 months. Over 3 years, net savings exceed \$927K. hyper2kvm handles the conversion from AWS AMI/VMDK, Azure VHD, or GCP images with automatic cloud agent cleanup, driver

replacement, and deployment. Keep bursty workloads in cloud, bring steady-state home.